

Customer Behavior Analysis

Professional Internship Project Report

Organization: Alfido Tech

Prepared By: DEVESH PANDEY

Table of Contents

1. Introduction
2. Objectives
3. Dataset Overview
4. Tools & Technologies
5. Data Cleaning
6. Feature Engineering
7. Exploratory Data Analysis
8. Customer Segmentation (RFM & K-Means)
9. Churn Analysis
10. Key Findings
11. Business Recommendations
12. Conclusion

Introduction

Customer behavior analysis helps organizations understand purchasing patterns, customer preferences, and retention. This project analyzes an e-commerce dataset containing approximately 250,000 customer transaction records using Python.

Objectives

- Clean and preprocess customer data.
- Perform feature engineering.
- Explore purchasing behavior through visualizations.
- Segment customers using RFM Analysis and K-Means.
- Analyze churn and provide actionable recommendations.

Dataset Overview

The dataset contains customer demographics, purchase history, product information, payment methods, returns, and churn indicators. After preprocessing, additional features such as Month, Year, Day, and Weekend were created.

Tools & Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Google Colab.

Data Cleaning

The dataset was inspected for missing values, duplicate records, and incorrect data types. Purchase Date was converted into datetime format and missing values were handled appropriately.

Exploratory Data Analysis

EDA included customer age distribution, gender distribution, product category analysis, payment method analysis, purchase amount distribution, monthly revenue trends, return analysis, churn distribution, top customers, and correlation heatmap. These visualizations highlighted customer purchasing patterns and business trends.

Customer Segmentation

RFM (Recency, Frequency and Monetary) analysis was performed to measure customer value. K-Means clustering grouped customers into high-value, loyal, regular, and inactive customer segments.

Churn Analysis

Customer churn was analyzed across payment methods and customer groups. The analysis helps identify customers requiring retention campaigns.

Key Findings

- High-value customers generate a significant share of revenue.
- Purchase frequency is strongly associated with customer value.
- Customer segmentation supports targeted marketing.
- Purchase trends vary across categories and payment methods.

Business Recommendations

1. Launch loyalty rewards for premium customers.
2. Re-engage inactive customers with personalized offers.
3. Promote top-performing product categories.
4. Improve product descriptions to reduce returns.
5. Monitor churn monthly.
6. Use segmentation for personalized campaigns.
7. Encourage repeat purchases through reward points.
8. Build dashboards for continuous monitoring.

Conclusion

The project successfully demonstrates the complete customer behavior analysis workflow, including preprocessing, visualization, customer segmentation, and business recommendations. The findings can help improve customer engagement, retention, and business decision-making.

References

Python Documentation, Pandas Documentation, Scikit-learn Documentation, Matplotlib Documentation, Seaborn Documentation.